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Hilti Fleet Management (A): Turning a Successful Business Model on Its Head

Introduction

In 1999 Marco Meyrat, General Manager of Hilti in Switzerland, received a visit from one of his key account managers, who told him, “I have good news and bad news.” The good news was that Hilti had just landed a major new contract from Batigroup, then the largest construction company in Switzerland. The bad news was that they’d have to reinvent their way of doing business to meet Batigroup’s requirements. This was especially tough because Hilti had been very successful with its extant business model. In 1999, Hilti was Liechtenstein’s largest manufacturing company and one of the largest power tool producers in the premium segment globally. With its high level of innovation, long-term vision, and superior product quality, Hilti was in some respects characteristic of the whole region, which encompassed Germany, Austria, and Switzerland.

Over the next few days, Marco Meyrat visited Batigroup and some of its construction sites. He saw how managing hundreds of power tools had put significant strain on Batigroup’s operations. Batigroup was a loyal and strategically important customer for Hilti and had requested a holistic tool management system. Its management wanted to outsource all processes related to the management of their tool fleet, including administrative activities, maintenance, and financing. Marco Meyrat saw a major opportunity in this request. In the past, his salesmen’s power tool demonstrations had led to the sale of one or two power tools per call – never more than that. Suddenly the opportunity had arisen to sell contracts rather than individual tools. A single contract might result in Hilti’s provision of 100, 200, or 300 tools to the customer. Meyrat reflected that a scalable business model built on this foundation could result in more than a simple key account project for Batigroup.

At about the same time, Pius Baschera, CEO of Hilti, was on one of his visits to the Swiss market organization. On his yearly visits to the various market organizations, he placed high value on personally strengthening one of Hilti’s core traits: closeness to the customer. He accompanied salesmen on their visits to customers and talked to the customers about their needs. Since he had once been General Manager of Hilti Switzerland, he was particularly interested in the Swiss market and keen on following its recent developments, challenges, and trends. Baschera had also heavily strengthened

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Hilti's firm performance. Since he took over the firm in 1994, Hilti's sales had exceeded \$2 billion USD for the first time, and profit had increased at a rate of almost 8%. Given his high performance demands and close relationships with customers, Baschera was certainly a tough man to impress with a new business idea.

Nevertheless, Baschera's visit gave Meyrat the perfect opportunity to arrange another business trip to Batigroup. He wanted to let Baschera form his own opinion on the matter. After their joint visit, which included meetings with the CEO and CFO of Batigroup, Baschera and Meyrat drove back to Hilti's headquarters in Schaan, Liechtenstein, discussing their latest observations on the way. Baschera confirmed that he saw a fascinating opportunity in Batigroup. He said, "One of the biggest challenges, even at the top management level, will be figuring out how to manage thousands of power tools."

Meyrat, in return, shared his belief that taking Batigroup's request and developing it further could serve as a useful way to differentiate Hilti from its competitors. "I see a clear customer need," he said. "The customer needs to organize all these small tools. We're talking about hundreds and thousands of tools that Hilti could manage. We could target a much higher share of wallet than what we currently have."

Baschera was impressed with Meyrat's enthusiasm. He said, "Marco, I think we really have something here. But we need to take a close look at the specific solution: the operating model, our competitive environment, and so on. Batigroup's request implies a radical break from Hilti's current business methods. Those same methods helped us become a leader in the premium power tools segment. I think we should handle this opportunity low profile."

Meyrat understood Baschera's point about starting on a small basis, but emphasized the scope of the opportunity, which he saw as more than a simple key account project. Baschera pressed him on this matter. "Why exactly do you think Hilti needs to shift in that direction?" Baschera asked. "We're on a successful path already. Our competitors are lagging behind."

Meyrat replied, "That may be true, but customer needs are starting to change. We are losing ground to competitors in the small-tool market. As you know, we are a premium brand, and in the segment of small tools, premium is of less value. Differentiation is hard for us to attain. We have to find new solutions, new customers--otherwise our brand will fade out in Switzerland."

Baschera had taken note of those recent developments. Still, he challenged Meyrat, saying, "I am a bit concerned about how far we can stretch this opportunity. There's still no clear urgency for switching away from our tried-and-true methods. Such a shift represents a high risk for Hilti, even in regard to initiating a small key account project for Batigroup. In the end, what would you expect the market size to be for such a solution?"

After some consideration, Meyrat replied, "I'm not sure. Business model innovation doesn't happen as often as product innovation. For Hilti, the chance to develop a new business model comes maybe two or three times per century. So we're dealing with high risks in regard to the market and we have no classic procedures at hand. Batigroup is handing us this opportunity on a silver platter. Honestly, I think our biggest concern should be seizing and benefiting from it before a competitor does so instead."

Baschera answered quickly, as they were now approaching Hilti's headquarters. "Marco," he said, "Hilti supports strategic and innovative initiatives, but whatever we do will be closely observed by the rest of the company."

The History of Hilti

When Hilti started as a family-owned business in 1941, it focused on contract manufacturing of construction tools. The scope, size, and reach of the company expanded significantly over the years. With its expansion, Hilti's goal was to concentrate on areas where it could achieve the leading position, and to grow faster than the market. In this vein, the firm continued to focus primarily on the commercial segment, rather than the segment of do-it-yourself customers.¹

Several distinctive features contributed to the success of Hilti's business model and differentiated the company from competitors such as Bosch, Makita, and Hitachi Power Tools. One such feature was the direct selling approach, which the company maintained since its foundation. Nearly two-thirds of all Hilti employees contacted customers regularly. They had approximately 250,000 daily contacts around the world.²

To best target customers with the direct selling approach, trade (e.g., building construction) and regional segments were meticulously separated according to their turnover and the potential they offered.³ Within these customer segments, Hilti sales employees tried to reach as many people as possible who were involved in tool-purchasing decisions (e.g., purchasers, users, foremen and tradesmen on building sites, specifiers, architects, and executives). The company also adopted a multi-channel approach to sales that included field workers, telephone support, Hilti centers, and an internet portal. It even sent videographers to worksites to observe how customers conducted daily tasks and used its products.

Hilti's customer-centric approach extended throughout the company. Even senior management spent some time with customers on-site. Most importantly, Hilti's R&D department was one of the first to adopt a Lead User concept, which integrated customers deeply in the process of developing new tools. Regular collaboration with customers and quality management enabled expedited development process times of about 24 months, in contrast to the average time of 3 to 3.5 years reported by industry observers.⁴ Moreover, Hilti was able to improve the productivity of its R&D department. Pius Baschera, then the CEO of Hilti, said, "The research centers of our competitors consist of 1,200 researchers or even more. So you might ask: 'How can Hilti compete against this work force?' Hilti is able to achieve the same output with a lot fewer researchers because of its close and permanent customer interaction. Every innovation is born out of customer needs, not just developed through over-engineering."

Hilti created the market for premium power tools and strove to position itself as a premium supplier. The median price of Hilti tools was above that of competitors, as the company focused on qualitative value-added products rather than short-term price benefits. Since it had to offer superior technology to compensate for charging a premium price, Hilti spent approximately 4.5% of its sales revenue on R&D.⁵

In the 1990s, Hilti began to branch out from product centrality while continuing to develop new appliances and technological improvements. In 1992, the company introduced warranty certificates; in 1996, it introduced three-year full service for breakers, a form of power tool; and in 1997, it introduced a six-month repair guarantee. Later, the value proposition of "Life Time Service" was included in the tool price, integrating features such as a lifetime repair cost limit, up to two years no costs, and a lifetime manufacturer's warranty.⁶

Companies like Hilti that focused on heavier power tools, priced around \$800, had smaller portfolios, consisting of about 50-100 different products (plus the corresponding variants). In order to compete with the broad portfolios of other companies that were three to four times larger, they now

sought to expand, offering light tools and smaller devices down to about \$200, as well as common consumables and accessories for only a few dollars.⁷ It became more difficult to differentiate with such offerings. For basic tools like angle grinders, there was no motive to compete on special features or higher performance, and no reason for customers to purchase them from a premium brand. Consequently, there was no need to demonstrate products by the means of a well-trained sales force. However, Hilti's sales force was built on the philosophy that there was a strong need to demonstrate the technological superiority, innovative features, and performance of Hilti products in live demonstrations. Such costly demonstrations were pointless for commoditized products, but in the late 1990s, firms like Hilti faced increasing commoditization. Marco Meyrat, Hilti's General Manager, said, "We found ourselves losing ground to our competitors in the small-tool market. [...] We are a premium brand, and in that segment premium was of less value. Differentiation was also harder for us to attain."⁸

The market analyses carried out during those years foresaw an intensification of competition. Analysts at Hilti predicted that rivals would grow in size, increasing their M&A activities and striving to gain higher market shares. Indeed, major competitors strove to expand their product portfolios and optimize their quantity/cost ratios.⁹ Additionally, the M&A activities of competitors increased, and they grew substantially. Such growth threatened Hilti by causing a scaling effect. Larger companies such as Bosch benefited from increased marketing and R&D budgets, additional sales channels, and improved production cost profiles.¹⁰ (See **Exhibit 1 a-c** for detailed information on Hilti's financial data and history.)

The Power Tools Industry circa the Year 2000

Power tools were used in a broad array of commercial and industrial applications, most significantly in construction and remodeling. Within those fields, power tools were used for new construction, repair, and maintenance of structures.¹¹ They were essential to many steps at construction sites, from drilling and demolition to cutting and grinding, fastening and installation, and firestop and insulation.¹² Firms also began to offer tools for such applications as engineering services, software, and value-adding services (e.g., lifetime services or guarantees). In looking for firms that might benefit from fleet management, three strategic groups emerged, all dedicated to supplying a construction site along this chain. These groups consisted of 1) sole producers of power tools, 2) application experts, and 3) full service providers.¹³ (See **Exhibit 2 a-b** for a further depiction of these three strategic groups, which are described below.)

Producers of Power Tools

Producers of power tools were firms that designed and manufactured power tools in-house. Although power tool technology was quite mature, these firms increasingly based their aggressive competition largely on products. Their intensive focus on R&D (average R&D expenditure ca. 4.5%) resulted in increased tool performance, additional features, and better ergonomics.¹⁴ Competition was also based on the scope of a company's portfolio and accessories. Firms had to embrace an entire portfolio of power tools in order to best satisfy the needs of customers. Portfolios spanned a wide range of product categories and performance levels, accessories and consumables, covering every step of every process requiring power tools at construction sites.¹⁵

There were hundreds of individual power tool producers around the globe. A cluster of small, niche power tool manufacturers in the area of Stuttgart, Germany alone included brands such as Metabo, Festool, Mafell, AEG Electric Tools, and C. & E. Fein (revenues less than 300 million Euros). The market

was fierce. From 2000 to 2001, the sales value of power tools decreased by 3%, and global sales of power tools stagnated.¹⁶ These changes resulted in increasing price competition within the already highly cost-driven industry, which was marked by low margins and high inventory turnovers.¹⁷ A relatively small number of major players began to dominate the global market. The major players were as follows: Black & Decker (U.S.), Bosch (Germany), Techtronic (Hong Kong), Hitachi (Japan), Makita (Japan), Illinois Tool Works (U.S.), and Hilti (Liechtenstein). (See **Exhibit 3 a-d** for a detailed analysis of the direct competitors and a market analysis of the power tools sector.)

The major players achieved a market share of more than 60% in 1999-2000 in the Hilti-relevant professional power tools market.¹⁸ Roughly half of the consolidation was driven by mergers and acquisitions (M&A). Techtronic was an exemplary case in this regard, also demonstrating that the number of Chinese companies rose significantly. The Asian construction industry showed increased electric drill demand, and growth rates in the Asian Pacific area were high (see **Exhibit 3c**). Firms such as Techtronic were eager to internationalize and to broaden their brand recognition by exporting to markets in developed nations. For large Western companies, too, the industry became highly globalized. Easy access to fast and worldwide logistics services, higher geographic dispersion of the customer base in the commercial sector,¹⁹ and decelerating growth rates in developed countries all contributed to globalization. Moreover, due to increasing price competition, larger power tool suppliers began investing directly in offshore production and foreign distribution facilities. In 1999, more than 50 nations possessed at least limited manufacturing capability for power tools.²⁰ Thus, these companies were eager to internationalize their activities to emerging markets such as Brazil, China, and Eastern Europe.

Application Experts

Alongside the power tool manufacturers, application experts were important suppliers in the construction business. Application experts primarily focused on the fastening and installation process step at construction sites. They were deeply involved in application processes and therefore mastered and dominated this decisive step. Application processes related to the upkeep and repair of sanitation, heating, ventilation, and air conditioning systems, as well as electrical and building installations, facade systems, fastenings for heat composite systems, construction and building chemistry (e.g., sealants, adhesives, and fire protection), and direct mounting. Application experts typically had an army of specialists and excellent R&D departments. They maintained direct contact with customers and had an in-depth knowledge of highly technological requirements in new buildings.

Hilti played in the field of applications experts, as did several competitors. Fischerwerke, a family-owned, medium-sized firm in Germany, was an exemplary company in this regard. Artur Fischer founded the firm with the legendary invention of the nylon anchor. Fischerwerke's core focus and special competence was the business segment for systems repair. It offered over 14,000 products and provided solutions for almost every repair problem. The firm targeted a wide range of customers, from do-it-yourself customers and skilled trade workers to key account partners. As a result, it understood nearly all applications at construction sites.²¹

Some producers of power tools, such as ITW, competed with Fischerwerke in this realm. The firm not only produced power tools, but also developed applications and repair systems. In this manner, ITW achieved a high level of application knowledge, which included both knowledge of a product and knowledge of its application at the worksite. Similar to Fischerwerke, ITW combined its products with extensive advice to customers and custom-made designs.²² Both firms were characterized by their close contact with customers for key account projects through their sales force.

Pure Service Providers and Retailers

Pure service providers and retailers were both trading companies, rather than original equipment manufacturing firms. They offered complete product portfolios, services, and solutions for construction jobsites, for processes ranging from measuring and aligning to drilling and demolition, cutting and grinding, fastening and installation, and firestop and insulation. They were generally big in size but did not manufacture products or do any significant R&D. Instead, they competed on the basis of convenient ordering and delivery services for an ever-wider offering of products at the jobsite. Their core capabilities included the sourcing of machinery, tools, and equipment from various manufacturers and the distribution of those items to construction companies and sites.²³

Würth was one of the major firms competing in this environment. Its strategy was to gain substantial market share in the construction business through "owning the job site." It had a dense network of stores to ensure high local proximity to the customer. Würth began to imitate Hilti's direct selling model and employed a direct sales force that was larger than Hilti's. It offered a nearly exhaustive product portfolio that ranged from power tools to consumables and even helmets, containing almost 200,000 products in total. However, Würth only manufactured 5% of what it sold.²⁴ The firm enhanced its offerings by increasingly providing after-sales services and insurance for stolen equipment.

Pure product providers were mainly the large and small retailers for power tools and construction equipment. Most power tool manufacturers selected these retailers as their main distribution channel. Retailers like Home Depot and Lowe's primarily targeted the general market of home improvement, and therefore both B2B firms (e.g., construction companies) and B2C markets. Both Lowe's and Home Depot were giants, achieving revenues of \$18.7 billion and \$45.7 billion, respectively, in 2000.^{25,26} Moreover, Home Depot announced that it wanted to accelerate the roll-out of professional customer programs, focusing its resources on its fastest growing segment. In European countries there were many smaller domestic retailers, such as OBI and Toom in Germany or Landi in Switzerland, not comparable to the large U.S. chains. In developing countries, the majority of purchases in the power tool industry were made through small retailers.²⁷ None of these retailers provided significant service offerings such as repairs or training.

Latent Customer Needs

In 2000, commercial users accounted for 70% of power tool sales.²⁸ This customer group demanded multi-functional products with special features. They wanted products that could be used for a variety of applications, such as hammer drills. These customers sought quality, durability, and ease of use, and were generally willing to make higher investments in exchange for longer product life and low maintenance cost. The residential B2C market, representing the other 30% of the market volume for power tools, was comprised of do-it-yourself customers. These customers had lower usage rates and less interest in new technologies. They based their purchases more on price and less on high performance features.²⁹

Generally, professional customers (i.e., construction companies) paid little attention to the purchase of power tools, even regarding them as c-parts. On average, the budget split for power tools was approximately 2% for professional customers. While firms like Caterpillar offered large and complex machinery worth millions of dollars to reasonably-sized construction sites, these sites only comprised some \$40,000 worth of power tool sales (\$400 per tool on average, including heavier tools around \$800 and lightweight tools and commoditized products around \$100; see **Exhibit 6a** for an exemplary customer with a tool park of around 100 power tools). Very large customers, like Batigroup in

Switzerland, owned tool fleets consisting of around 600 power tools. In 2000, Hilti provided 37% of the tools for Batigroup, and all other tools were bought from different manufacturers.³⁰ Since Hilti provided Batigroup with premium heavyweight tools, which cost more than the average, it achieved a share of customer wallet of around 50%.³¹ Personnel and materials proved to be bigger cost blocks for customers and, most importantly, larger equipment received more attention from construction companies. However, the 2% of the budget split only comprised the direct tool costs; higher performance of power tools could indirectly optimize efficiency and productivity.³²

Hilti's extant business model, in which firms owned their power tools, meant that construction companies were responsible for all activities related to the management of tools. However, the construction site managers were busy with their core business. Unpredictable bills and administrative processes due to repairs and replacement of tools interrupted the work and diverted the focus away from core tasks at the construction site. Construction companies were also ineffective at managing power tools, which delayed projects significantly.

Overall, much of the cost related to power tools occurred because of the repair process. Costs for a single repair could add up to 50% of the price of new tools.³³ Moreover, broken drilling machines caused heavy losses in the construction business, since workers could not carry out their work without those machines. In the case of repairs, service parts and providers were often not quickly reachable. The construction site manager had to return the tool to the technical center or central warehouse of the construction company for damage assessment, after which the construction company would either repair the tool or send it to the power tool supplier for repair. The average repair time was up to 11 days and tools were often unavailable when most needed.³⁴ Repairs also involved several costly logistical and administrative steps³⁵ (see **Exhibit 4** for an exemplary repair process). Consequently, many days were wasted due to broken equipment.

A customer's inability to manage its tools properly resulted in other significant hidden costs. Customers often clung to their products for too long, and spent more money on repairs than they would have paid for a new product. Moreover, power tools showed efficiency losses if they were not maintained regularly. A worn-out tool reduced productivity drastically. A typical example of a small 10-man business showed that efficiency losses could lead to monthly losses of over \$700.³⁶ Perhaps as a result of power tool commoditization, construction workers viewed these tools as virtually disposable.³⁷ Tools were often stolen at construction sites, as they were not stored properly. Frequently, the construction sites did not differentiate between losses and thefts. In a 5-year period a loss of around 15-25% of tools was common.³⁸

A precise tabulation of the hidden costs related to power tools in a construction firm generally did not appear in any accountancy. However, these hidden costs significantly exceeded the investment and regular maintenance costs of a tool. On average, customers had about six times higher costs with regards to a power tool when taking into account the health and safety risks, opportunity costs due to neglect of novel technologies, and fragmented and unstructured tool management.³⁹

Hilti also observed that its customers used power tools of many different brands. This meant increased costs as the companies found it difficult to standardize the repair and maintenance processes. At the same time, the profusion of cheap, battery-powered hand tools left worksite tool tables scattered with mismatched components from different manufacturers.⁴⁰

Managing, maintaining, and transporting hundreds or even thousands of small tools was thus burdensome to customers. Construction sites struggled to manage their fleet of tools over the course of a project. Moreover, specialized drilling machines were very costly. Pius Baschera, then-CEO of Hilti, said, "Tool management is a pain. A construction crew's purpose is to build a house, not manage

tools.”⁴¹ Having considered this problem, Hilti executives realized that delivering convenience and establishing customization were the future success factors in the power tools business.

The Emergence of a New Business Model

In 2000, Meyrat and his team responded to Batigroup with an innovative business model that incorporated fleet management. Instead of buying tools as before, Hilti customers could purchase permanent “tool availability,” which delegated to Hilti the responsibility of supplying, repairing, and replacing tools, as well as providing coverage in the event of theft. Hilti tool management also covered temporary peak demand or the punctual need for specialized tools. Hilti expected this value proposition to lead to increased performance for customers due to less downtime, and to greater safety, since Hilti would provide modern, perfectly maintained, and state-of-the-art tools. More specifically, the value proposition included the following aspects (see **Exhibit 5** for further information on the initial value proposition):

- Full financing of the fleet
- Monthly billing, same price guaranteed
- Free maintenance and calibration of tools
- Free loaned machines during repair downtimes
- Free tool collection and delivery services
- Provision of temporary power tools at a daily rate exclusive for Fleet Management customers (e.g., in the event of peaks or for tools that were not part of the fleet)
- Provisions for theft: Coverage of 80% of the outstanding monthly rates, while the customer had a deductible of 20%. (Prerequisite was that the customer must provide a police report.)
- An online tracking system and comprehensive inventory management
- Renewal and recycling of tools at the end of the defined usage time (typically a 2-5 year period)

The team analyzed how a fleet management system with a high customer penetration rate of Hilti tools (100% share of customer wallet) could positively impact Batigroup’s tool park. They concluded that Batigroup’s tool park could be reduced by 7% in a fleet management contract. Moreover, the higher the share of customer wallet, the higher the tool exchange rate offered by Hilti, and the lower the average tool age and tool exchange age.⁴²

In the past, Hilti had sold tools, spare parts, and maintenance services directly. Thus, the revenue model had to be completely redefined for the new business model. The new model replaced large lump sums of income with regular smaller payments, and assets disappeared from customers’ balance sheets. The basic structure of leasing contracts was readily adapted from the automotive industry, as the customer had monthly expense payments instead of full upfront investments.⁴³ (See **Exhibit 6 a-e** for an exemplary calculation illustrating how fleet management compares to traditional sales methods.)

Exhibit 1a Hilti Financial Highlights 1997-2000

<i>Year</i>	2000 <i>(in million CHF)</i>	<i>(in million USD)</i>	1999		1998		1997	
<i>Annual results</i>								
Net sales	3138.0	1945.6	2814.0	1744.7	2630.0	1630.6	2580.0	1599.6
Depreciations	141.5	87.7	139.2	86.3	139.8	86.7	136.4	84.6
Operating result	309.0	191.6	260.5	161.5	226.1	140.2	193.6	120.0
Financial result	102.5	63.6	103.4	64.1	69.2	42.9	55.1	34.2
Net profit	368.8	228.7	321.3	199.2	262.2	162.6	225.7	139.9
Cashflow	510.3	316.4	460.5	285.5	400.6	248.4	362.1	224.5
Return		0.0		0.0		0.0		0.0
(-)operating capital in %	15.1	9.4	13.0	8.1	11.7	7.3	9.7	6.0
(-)total capital (ROI) in %	11.2	6.9	10.1	6.3	9.4	5.8	8.3	5.1
(-)equity capital in %	15.8	9.8	13.8	8.6	12.5	7.8	11.7	7.3
<i>Balance sheet</i>								
Equity capital	2202.0	1365.2	2470.0	1531.4	2197.0	1362.1	2013.0	1248.1
(-) in % total capital	59.0	36.6	65.0	40.3	64.0	39.7	63.0	39.1
Long term borrowed capital	849.0	526.4	694.0	430.3	604.0	374.5	567.0	351.5
Short term borrowed capital	706.0	437.7	655.0	406.1	613.0	380.1	607.0	376.3
Investments in intangible assets and property, plant and equipment	168.0	104.2	152.0	94.2	156.0	96.7	138.0	85.6
Tangible and intangible assets	675.0	418.5	684.0	424.1	653.0	404.9	653.0	404.9
Financial assets	1789.0	1109.2	1484.0	920.1	1519.0	941.8	1266.0	784.9
Working capital	1293.0	801.7	1651.0	1023.6	1242.0	770.0	1275.0	790.5
Balance sheet total	3757.0	2329.3	3819.0	2367.8	3414.0	2116.7	3194.0	1980.3
Total dividend	98.8	61.3	86.2	53.4	74.9	46.4	63.4	39.3
<i>Work force (Annual average)</i>								
Number of employees	13780		13190		12700		12110	
USD-CHF	1.611		1.5907		1.3735		1.4633	
CHF-USD ^a	0.6207		0.6287		0.7281		0.6834	

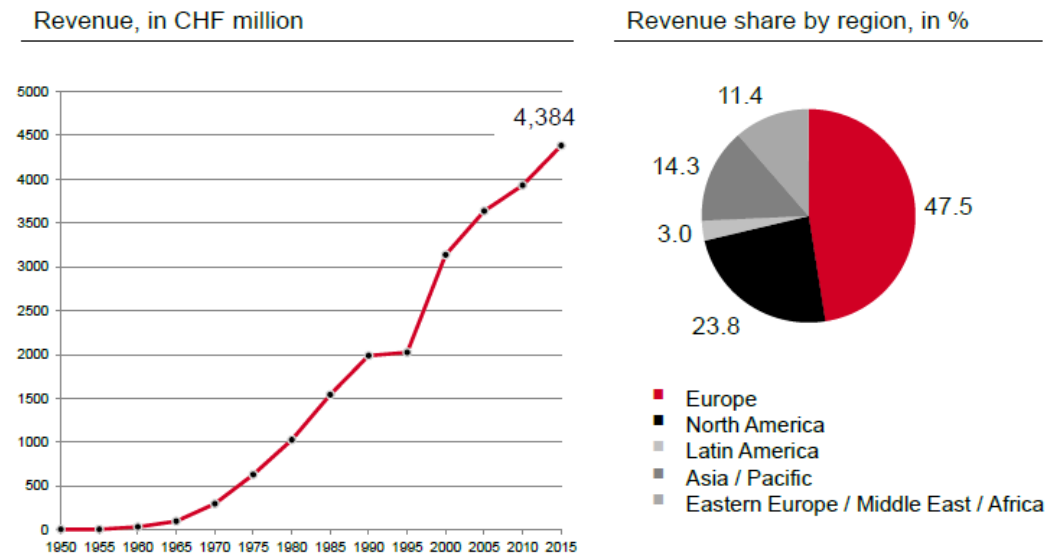
Source: Hilti, 2000 Annual Report (Schaan, Liechtenstein: Hilti, 2001).

^a Exchange rate of 2000 for all years (source: boerse.de).

Exhibit 1b Hilti Financial Highlights 1997-2000 per Market

	2000		1999		1998		1997	
<i>Net sales by region</i>	<i>(in million CHF)</i>	<i>(in million USD)</i>						
Europe/Africa	1844	1143.28	1748	1083.76	1672	1036.64	1575	976.5
North America	833	516.46	677	419.74	585	362.7	545	337.9
Latin America	95	58.9	80	49.6	91	56.42	80	49.6
Asia/Pacific	366	226.92	309	191.58	282	174.84	380	235.6
<i>Expenditure for securing future</i>								
	<i>(in million CHF)</i>	<i>(in million USD)</i>						
Education and training	36	22.32	31	19.22	24	14.88	23	14.26
Research and development	138	85.56	123	76.26	96	59.52	82	50.84
Investments in tangible and intangible assets	168	104.16	152	94.24	156	96.72	138	85.56
<i>Worldwide workforce (Annual average)</i>								
Europe/Africa	9170		8800		8424		8133	
North America	2070		1955		1871		1854	
Latin America	760		770		707		589	
Asia/Pacific	1780		1665		1698		1534	

Source: Hilti, 2000 Annual Report (Schaan, Liechtenstein: Hilti, 2001).

Exhibit 1c Hilti Revenues 1950-2015

Source: "Hilti ON!Track: with Transparency comes Efficiency," Hilti internal presentation on fleet management, August 30, 2016, "Digitalisierung in der Industrie" Conference, Rapperswil, CH, slide 4.

Exhibit 2a Hilti's and Power Tool Producers' Portfolios along the Process Steps at a Construction Site



Source: "Hilti ON!Track: with Transparency comes Efficiency," Hilti internal presentation on fleet management, August 30, 2016, "Digitalisierung in der Industrie" Conference, Rapperswil, CH, slide 6.

Exhibit 2b Comparison of Application Experts and Full Service Providers/Retailers with Producers of Power Tools along the Process Steps at Working Sites

Process steps at working site:

	Engineering and Manuf.	Measuring and aligning	Drilling and demolition	Cutting and grinding	Fastening and installation	Firestop and insulation	Service and Software
Producers of Power Tools*	✓	✓	✓	✓	(✓)**	(✓)**	✓
Application Experts*	✓	✗	✗	✗	✓	✓	✓
Full Service Providers and Retailers*	✗	✓	✓	✓	✓	✓	(✓)***

*Producers of Power Tools and Application Experts could be referred to as "Original Equipment Manufacturers". The Full Service Providers and Retailers could be referred to as "Trading Companies"

** Not all Producers of Power Tools offered products for these process steps. ITW and Hilti did offer products for these process steps.

*** Full Service Providers did offer services (e.g., guarantees, insurance options, etc.), retailers did not.

Source: Compiled by casewriter based on interview with Pius Baschera, August 30, 2016, and Pius Baschera, "Hilti Case Study: Strategic Management in Practice," Swiss Federal Institute of Technology (Zurich: November 2015).

Exhibit 3a Data on Hilti's Direct Competitors

	Hitachi (Corporate parent)	Robert Bosch GmbH (Corporate parent)	Makita	Black&Decker (including Dewalt)	Techtronic Industries	
Origin	Japan	Germany	Japan	USA	Hong Kong/China	USA
Type	Conglomerate	Conglomerate	Power tools manufacturer	Power tools manufacturer	Power tools manufacturer	Global diversified manufacturer
Net Sales	75,483 million USD	29,999 million USD	1,312 million USD	4,560 million USD*	587 million USD (Power tools: 397 million USD)	9,983 million USD
Operating Income	1,645 million USD		82 million USD	503 million USD	38.5 million USD	1,563 million USD
Net Income	160 million USD	1,269 million USD	31 million USD	282 million USD	24.6 million USD	957 million USD
Employees	330,000	197,000	7915	23600	8000	55,000
Domestic			23.5%	(-)	(-> North America: 85%)	ca. 60%
Percentage of Power Tools Activities	0.80%	4.54%				
Subsidiary/Division	Hitachi-Koki	Consumer Goods and Building Technology				
Net Sales	1,208 million USD (Power tools: 617 million USD)	6,682 million USD (Power tools: 1,363 million USD)				
Operating Income	17.9 million USD (Power tools: 20 million USD)	(-)				
Net Income	24.4 million USD	(-)				
Employees	(-)	(-)				
Domestic	51%	13%				

**(70% of sales with power tools,
rest: accessories or fastening and
assembly systems)*

Exchange rate (2000): ¥123.90=U.S.\$1 2,122 USD = 1 DM 1USD = 0.47 DM HK7,7425=1USD (-)

Source: Data extracted from: Hitachi, 3/1999-3/2000 Annual Report (Tokyo: Hitachi, 2000); Bosch, 2000 Annual Report (Stuttgart, Germany: Bosch, 2001); Makita, Annual Report 2001-2002 (Aichi, Japan: Makita, 2002); Black & Decker, 2000 Annual Report (Baltimore, Maryland: Black & Decker, 2001); Techtronic Industries, 2000 Annual Report (Hong Kong: Techtronic Industries, 2001); Illinois Tool Works Inc., Annual Report 2000 (Glenview, Illinois: Illinois Tool Works, 2001). Average exchange rates for respective years acquired from: boerse.de, accessed October 2016.

Exhibit 3b Description of Hilti's Direct Competitors

Black & Decker. A U.S. based firm, Black & Decker was focused solely on manufacturing power tools and accessories. It served both B2C and B2B markets, the latter through its DeWALT brand. In 2000, it achieved net sales of approximately \$4.56 billion and employed 23,600 people. Black & Decker marketed its tools globally. It held manufacturing operations in ten countries, and its tools were distributed in more than 100 countries by intermediaries in the medium-priced segment. The firm focused on five key strategies: sales growth, world-class cost and quality, supply chain re-engineering, e-business leadership, and globalization. It generally worked with powerful retailers like Home Depot or Lowe's. To increase global sales growth, Black & Decker supported its key retail partners as they expanded in Latin America and elsewhere around the world, and made heavy investments in e-business. In the year 2000, it highly encouraged Six Sigma initiatives to improve the quality of its products. Black & Decker initially based the majority of its sales on do-it-yourself customers, but after acquiring the firm DeWALT, it added a large commercial sector to its portfolio and phased out its rather small initiatives in the professional sector, such as the Elu brand in Europe.

Both Black & Decker and DeWALT had strong brand recognition, and activities were consolidated within these brands. Sales of DeWALT tools and accessories grew to more than \$1.5 billion in 2000. DeWALT increasingly provided a comprehensive offering of differentiated and low-innovation solutions and commodities to suit customer needs. In the wake of Black & Decker's bolt-on acquisition strategy, which was closely aligned with DeWALT's growth plans, the firm entered into two important product categories in 2000 with the acquisition of Momentum Laser, Inc., a construction laser company, and Emglo Products, L.P., a manufacturer of air compressors. Despite its active acquisition strategy, Black & Decker generated \$1 billion in free cash flow between 1995 and 2000, and repaid nearly the same amount of debt during that period.

Bosch. A German technology conglomerate, Bosch had 197,000 employees and achieved net sales of around \$30 billion in the year 2000. Power tools activities contributed \$1.36 billion to its total sales. It served both the B2C and B2B markets. Bosch's major segments were automotive technology, industrial technology, and consumer goods and building technology. The power tools division was attached to the consumer goods and building segment. Bosch was one of the world's leading producers of power tools in the low-to-medium-priced segment. Its power tools were produced at 32 locations worldwide, and its marketing organization operated in 94 countries. Foreign sales accounted for 87% of total sales. Bosch offered the Bosch, Skil and Dremel brands. The Bosch brand showcased its commitment to both customer segments by introducing a green line of products for residential customers (B2C) and a blue line for commercial customers (B2B). Each line was distributed through both small and large retail chains.

Although the German market declined in value in the year 2000, sales in the power tool division achieved 15% growth. This was primarily because of the increasing demand in North America and Asia. Bosch's major focus was on continuous innovation and expansion of its portfolio. The firm carried a large range of power tools and announced numerous product launches. For instance, in 2000, Bosch was one of the first companies to present a comprehensive line of 24V batteries for power tools ranging from hammer drills to screwdrivers to hacksaws.

Techtronic. Techtronic was one of the fast growing major players in Hong Kong and China. It served both the B2B and B2C segments. Techtronic employed nearly 8,000 employees and grew to \$587 million in 2000. Power tools contributed \$397 million to that total. Techtronic was founded in 1985, and initially focused on "white label" production of power tools for Western firms. It provided much of Sears, Roebuck and Company's Craftsman line of power tools. Its portfolio also contained floor care

appliances and solar and electronic products, although they were a minor segment. By 1990 Techtronic was listed on the Hong Kong Stock Exchange. It had operations and production facilities around the world. In order to gain a more prominent standing worldwide and rapidly increase market share, the firm began offering competitively priced tools and engaged in several M&A activities from the late 1990s onward.

Techtronic enjoyed a compound annual growth rate of 27% between 1995 and 2000. In 2000, it acquired the Japanese Ryobi group, then the largest supplier of battery-powered tools. This acquisition positioned Techtronic as a leading company in the global power tool market. It laid the foundation for new alliances with some of the world's most powerful retailers. In 2000, Techtronic announced that it would extend its core businesses into Europe and expand its brand portfolio based on acquisitions. In 2001 it acquired Homelite, a U.S.-based manufacturer.

Hitachi. A leading Japanese technology conglomerate, Hitachi achieved net sales of approximately \$75.5 billion in the period between March 1999 and March 2000, of which \$617 million came from the power tools segment. Hitachi predominantly relied upon commissioned product sales by retailers in the B2B and B2C segments. Its subsidiary, Hitachi-Koki Co. Ltd., manufactured and sold products in three distinct markets: power tools, printing systems, and scientific instruments. Hitachi-Koki achieved net sales of \$1.2 billion, \$617 million of which came from the power tools segment. Hitachi's power tool activities focused mainly on domestic markets. It began to concentrate on new production bases in China in order to expand. Hitachi aimed to double production at its Chinese facilities, manufacturing cost-competitive power tools and cordless tools for professional and do-it-yourself users in markets worldwide. To achieve fast sales growth in North America, Hitachi targeted home center channels in that market. In 2000, the firm established its Home Center Sales Promotion Department to integrate all home center sales functions. It shifted away from commissioned product sales by retailers, toward a direct sales structure in the Tokyo and Yokohama metropolitan areas, where it established Hitachi Koki Sales Co., Ltd.

Makita. A Japanese premium producer of power tools, Makita designed, manufactured, and marketed an extensive lineup of power tools. The domestic market accounted for 23.5% of its sales. Makita achieved net sales of \$1.3 billion in 2000 and employed roughly 8,000 people. With its wide range of products, Makita was well positioned to meet the specific needs of individual European regions, such as the demand for environmentally friendly products in Northern Europe and large-model power hammers in Eastern Europe. Consequently, in 2000, Makita was able to sustain its sales in the Western European market at roughly the same level as in the previous year and to increase its net sales in the Eastern European market at a double-digit rate. Since the overall power tool market was not projected to grow substantially, Makita planned several strategic moves. First, in addition to the hardware stores that had previously been the focus of its marketing programs, Makita progressively directed its marketing activities at corporate customers, and began to strengthen its relationships with exclusive distributors and hardware stores that catered to professional craftspeople. Second, in 2001, to increase the appeal of Makita products to professional power tool users, product specialists were assigned to visit those users, explaining and demonstrating newly developed products. Third, Makita worked to expand its business in accessories, which had previously been relatively small in scale.

ITW (Illinois Tool Works). A U.S. firm, ITW achieved net sales of almost \$10 billion in 2000, with 55,000 employees. ITW marketed a very broad and highly engineered product line of power tools and fastening systems for the B2B segment. ITW served the automotive, construction, general industrial, and food retail and service markets. ITW designed and produced a variety of products. The field of engineered products, which included short lead-time plastic and metal components and fasteners,

comprised 31% of its net sales in the construction market. Specialty systems, which included longer lead-time machinery and related consumables, made up 9% of net construction sales, and equipment and consumable systems, which ranged from specialty exercise equipment to small electrical appliances, made up another 31%. ITW was a highly decentralized company with nearly 600 business units that were realized through continuous acquisitions and inorganic growth. The 45 acquisitions realized in 2000 added around \$1 billion in sales, most notably through Siddons Ramset Limited (construction) in Australia and A. J. Gerrard (industrial packaging) in the United States. ITW had a strong focus on product development to meet very specific customer needs. Its strategic motto was: “Build your business around the 20% of your customers who represent 80% of your revenues.”

Source: Data and information extracted from annual reports of respective firms: Black & Decker, 2000 Annual Report (Baltimore, Maryland: Black & Decker, 2001); Bosch, 2000 Annual Report (Stuttgart, Germany: Bosch, 2001); Techtronic Industries, 2000 Annual Report (Hong Kong: Techtronic Industries, 2001); Hitachi, 3/1999-3/2000 Annual Report (Tokyo: Hitachi, 2000); Makita, Annual Report 2001-2002 (Aichi, Japan: Makita, 2002); Illinois Tool Works Inc., Annual Report 2000 (Glenview, Illinois: Illinois Tool Works, 2001).

Exhibit 3c Market Data: World Power Tool Demand by Regions & Countries

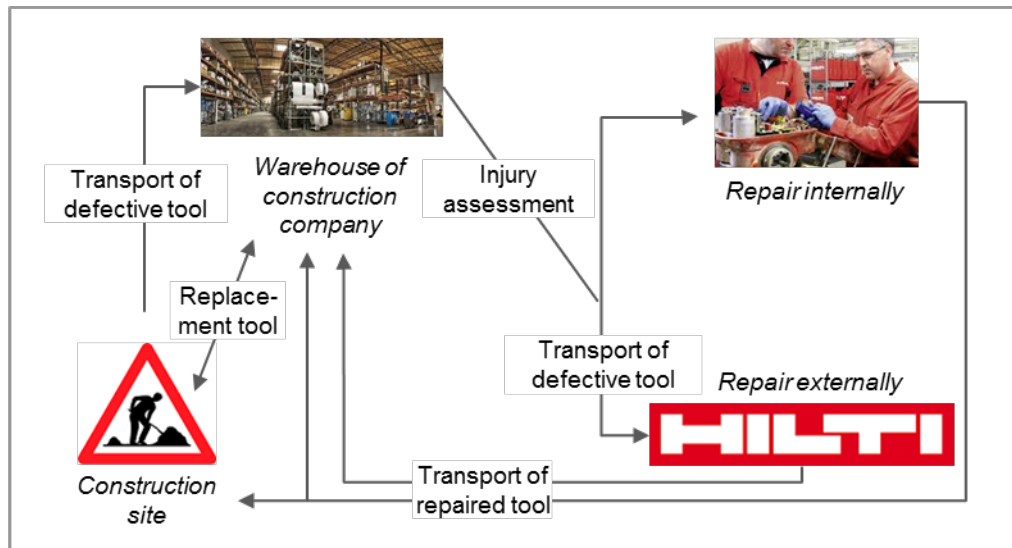
	1999	2004	2004/1999 Annual Growth Rate (%)
Gross Fixed Investment (bil.2000 USD)	9,613	12,040	4.9
USD Power Tools/1,000 USD GFI	2	1.9	(-)
World Power Tool Demand (mil current USD)	18,995	22,800	5
<i>North America:</i>	7,550	9,150	4.5
United States	6,550	1,960	4.4
Canada & Mexico	1,000	1,190	4.9
<i>Western Europe:</i>	5,580	6,260	4.1
<i>Asia/Pacific:</i>	4,005	5,005	6.4
China	510	885	9.4
Japan	1,620	1,710	3.7
Other Asia/Pacific	1,875	2,410	7
<i>Other Regions:</i>	1,860	2,385	6.4
Latin America	560	690	7.3
Eastern Europe	750	970	5.8
Africa/Mideast	550	725	6.5

Source: Michael Dennen and Andrew Gross, "The global market for power tools - focus on industry and markets," *Business Economics*, July 2006, p.66-73.

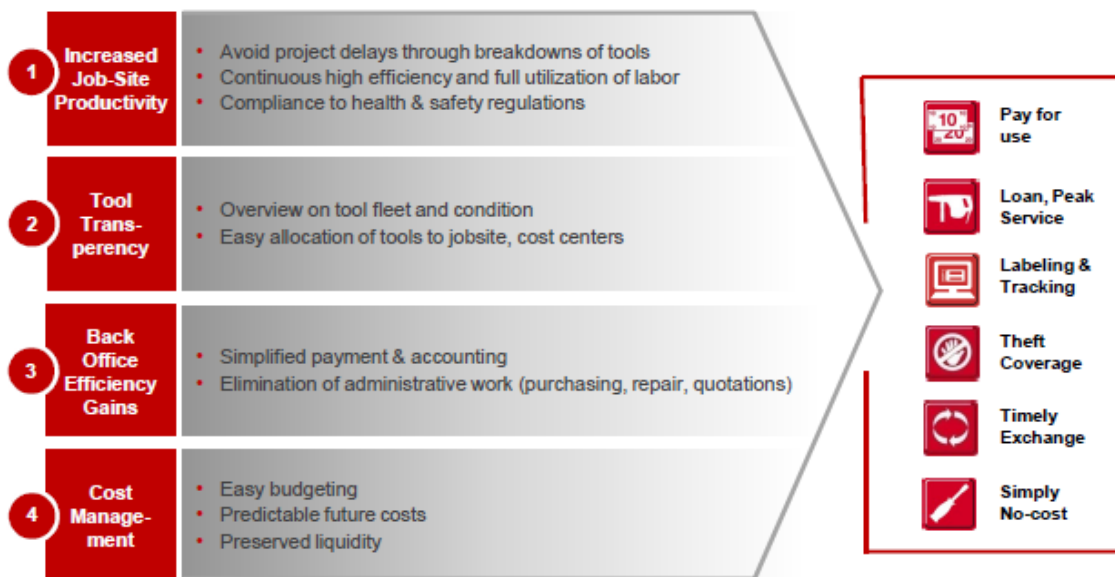
Exhibit 3d Market Data: World Power Tool Demand by Product, Power Source, & Market

	1999	2004	2004/1999 Annual Growth Rate (%)
Gross Fixed Investment (bil.2000 USD)	9,613	12,040	4.9
USD Power Tools/1,000 USD GFI	2	1.9	(-)
World Power Tool Demand (mil current USD)	18,995	22,800	5
<i>Power Tool Demand by Type:</i>			
Electric Tools	13,655	16,715	5.3
<i>By Product:</i>			
Drills	3,025	3,715	5.8
Saws	1,810	2,520	6.2
Other	3,985	4,675	5
Parts & Accessories	4,385	5,805	4.3
<i>By Power Source:</i>			
Plug-In	6,809	7,535	4
Cordless	2,011	3,375	8.7
Pneumatic Tools	3,407	3,890	4.4
Engine-Driven & Other	1,933	2,195	4.2
<i>Power Tool by Market:</i>			
Professional	13,825	16,275	5
Consumer	5,170	6,525	5.1

Source: Michael Dennen and Andrew Gross, "The global market for power tools - focus on industry and markets," *Business Economics*, July 2006, p.66-73.

Exhibit 4 Standard Procedure for Repairing a Broken Tool without Fleet Management

Source: Casewriter based on interview with Arndt Paul, June 13, 2016.

Exhibit 5 Hilti Fleet Management Value Proposition in 2001

Source: Company documents.

Exhibit 6a Exemplary Customer Data Specification I/IV

Exemplary Customer (around 8 average construction sites in parallel)	Number	Line No.
Current tool fleet	100	A1
Spare tools	4	A2
Total fleet	104	A3
Average age of tool fleet (in years)	2	A4
Lifetime of a tool in the fleet (in years)	4	A5
Rechargeable batteries per year	3	A6
Chargers per year	2	A7
Calibrations per year	10	A8
Thefts per year	3	A9
New purchases per year	26	A10
Repairs per year	15	A11
Downtimes per year	41	A12
Invoices per year	56	A13
Workers	35	A14
Work days per year	224	A15
Average tool usage per day in hours	1.5	A16

Source: Customer data specifications and calculation method are based on a Hilti customer presentation entitled: "Geräteparkanalyse: Bestandsanalyse und Optimierung Ihres Geräteparks."

Exhibit 6b Exemplary Customer Data Specification II/IV

Personnel, hardware and capital cost	in USD	Line No.
Average price of a Hilti tool in the fleet	400	B1
Personnel costs per device failure (downtime costs of workers and other employees)	17	B2
Personnel costs per tool purchasing process (for purchasing, administration, invoicing)	15	B3
Wage of a worker per hour	35	B4
Costs per rechargeable battery	65	B5
Costs per charger	65	B6
Costs per calibration	15	B7
Costs per repair (On average 50% of a tool price)	200	B8
Internal overhead/admin costs per invoice (e.g., repairs)	15	B9
Capital costs/Calculated interest rate in %*	2.5	B10
Average capital lock-up	20,800	B11
Assumptions:		
Linear depreciation, per year:	100	B12
Residual value of tools after 4 years	0	B13

Source: Customer data specifications and calculation method are based on a Hilti customer presentation entitled: "Geräteparkanalyse: Bestandsanalyse und Optimierung Ihres Geräteparks."

* Depending on customer and year – rough estimation.

Exhibit 6c Exemplary Customer Data Specification III/IV

Cost Reduction through Hilti Fleet Management	in %	Line No.
Hilti % Cost Reduction through less downtime	50	C1
Hilti % Cost Reduction through lower purchasing costs	80	C2
Hilti % Cost Reduction through fewer theft costs	80	C3
% Reduction of tool quantity through Hilti FM optimization	7	C4
Avoidances/Effects through Hilti Innovation		
% Loss of efficiency per year (avoided)	3	C5
% Increase in productivity (effect achieved)	3	C6
Hilti Fleet Management		
Avg. FM price per month for a tool in EUR (exemplary for this customer)	17.5	C7
Average margin per FM contract*	45	C8

Source: Customer data specifications and calculation method are based on a Hilti customer presentation entitled: "Geräteparkanalyse: Bestandsanalyse und Optimierung Ihres Geräteparks."

* This is a fictitious value and is not based on any Hilti data. The number has been included for teaching purposes.

Exhibit 6d Exemplary Customer Data Specification IV/IV

Value of Tool Fleet	in USD	Line No.		
Value of tool fleet at start of each year	26,000	D1		
Value of tool fleet at end of each year	15,600	D2		
Average value of tool fleet	20,800	D3		
Loss of efficiency in h for conventional tool fleet (without FM):	Tool efficiency in h	Line No.	Loss of efficiency in h	Line No.
Year 1 tools:	8736*	D4	0	D8
Year 2 tools:	8474**	D5	262	D9
Year 3 tools:	8220***	D6	254	D10
Year 4 tools:	7973****	D7	247	D11
Total loss of efficiency in a 4 year period			763	D12

Source: Customer data specifications and calculation method are based on a Hilti customer presentation entitled: "Geräteparkanalyse: Bestandsanalyse und Optimierung Ihres Geräteparks."

*=A10*A16*A15, **=D4-(C5/100)*D4, ***=D5-(C5/100)*D5, ****=D6-(C5/100)*D6.

Exhibit 6e Exemplary Customer Data Comparison of Fleet Management and Conventional Model

Conventional Model (old model)		Fleet Management	
Calculation based on Exhibit 11A-D		Calculation based on Exhibit 11A-D	Result (USD)
Direct costs for tool fleet per year for customers	Price for FM per year	0	$((100-C4)/100)*A3*12*C7$ 20,311
	Costs for new tools per year: $A10*B1$	10,400	0
	Depreciation costs for old tools: $(A3-A10)*B12$	7800	0
	Capital lock-up costs $D3*(B10/100)$	520	0
	Repair costs $(A11*B8)$	3,000	0
	Theft costs* $A9*(B1/2)$	600	$((100-C3)/100)*(A9*(B1/2))$ 120
	Calibration costs $A8*B7$	150	0
	Costs for complementarities $(A6*B5 + A7*B6)$	325	0
	Sum:	Direct costs for customers in conventional model 22,795	Direct costs in FM model 20,431
Additional (hidden) costs per year for customer	Costs for downtimes $(A12*B2)$	697	$((100-C1)/100)*(A12*B12)$ 348.5
	Administrative and purchasing costs $(A13*B3)$	840	$((100-C2)/100)*(A13*B3)$ 168
	Loss of efficiency $(D12*B4)$	26,701	0
	Productivity losses $(C6/100)*A16*A15*A14*B4$	12,348	0
	Sum:	Hidden costs for customers in conventional model 40,586.10	Hidden costs in FM model 516.50
Total Sum		Current "willingness to pay" per year 63,381.10	Total Costs of FM for customer per year 20,947.70

Source: Customer data specifications and calculation method are based on a Hilti customer presentation entitled: "Geräteparkanalyse: Bestandsanalyse und Optimierung Ihres Geräteparks."

* Theft costs were based on the average value of a power tool (considering depreciation), rather than the price of a new Hilti power tool.

Endnotes

¹ Hilti, Hilti.com, accessed July 2016.

² Interview with Pius Baschera, August 30, 2016.

³ Sven Reinecke, Jens Keller, and Marc Cristofolini, "Building Profitable Growth through 'Champion 3C' Strategy – The Case of Hilti," The Case Center No. 513-099-1 (The Case Center, 2013).

⁴ Sven Reinecke, Jens Keller, and Marc Cristofolini, "Building Profitable Growth through 'Champion 3C' Strategy – The Case of Hilti," The Case Center No. 513-099-1 (The Case Center, 2013).

⁵ Interview with Stefan Nöken, August 30, 2016.

⁶ Stefan Michel, "Hilti Fleet Management: Strategically Moving from Products to B2B Solutions," The Case Centre No. IMD-3-2354 (The Case Centre, 2013).

⁷ Interview with Arndt Paul, June 13, 2016; Interview with Jan Doongaji, on August 30th, 2016.

⁸ Mark W. Johnson, *Seizing the White Space – Business Model Innovation for Growth and Renewal* (Boston, MA: Harvard Business Press, 2010).

⁹ Pius Baschera, "Hilti Case Study: Strategic Management in Practice," Swiss Federal Institute of Technology (Zurich: November 2015).

¹⁰ Pius Baschera, "Hilti Case Study: Strategic Management in Practice," Swiss Federal Institute of Technology (Zurich: November 2015).

¹¹ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.

¹² "Hilti ON!Track: with Transparency comes Efficiency," Hilti internal presentation on fleet management, August 30, 2016, "Digitalisierung in der Industrie" Conference, Rapperswil, CH.

¹³ Interview with Pius Baschera, August 30, 2016.

¹⁴ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.

¹⁵ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.

¹⁶ "Elektrowerkzeuge – Weltmarkt rückläufig," press release, April 8, 2002, on Bosch Power Tools website, <http://www.bm-online.de/allgemein/weltmarkt-ruecklaeufig>, accessed December 2016.

¹⁷ Mark W. Johnson, *Seizing the White Space – Business Model Innovation for Growth and Renewal* (Boston, MA: Harvard Business Press, 2010).

¹⁸ Pius Baschera, "Hilti Case Study: Strategic Management in Practice," Swiss Federal Institute of Technology (Zurich: November 2015).

¹⁹ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.

²⁰ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.

²¹ Fischerwerke GmbH & Co KG, fischer.de, accessed October 2016.

²² Illinois Tool Works Inc., 2000 Annual Report (Glenview, Illinois: Illinois Tool Works, 2001).

²³ Interview with Pius Baschera (Hilti), August 30, 2016; Pius Baschera, "Hilti Case Study: Strategic Management in Practice," Swiss Federal Institute of Technology (Zurich: November 2015).

²⁴ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.

- ²⁵ Lowes, 2000 Annual Report (Wilkesboro, North Carolina: Lowes, 2001).
- ²⁶ The Home Depot, 2000 Annual Report (Atlanta, Georgia: Home Depot, 2001).
- ²⁷ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.
- ²⁸ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.
- ²⁹ Michael Dennen and Andrew Gross, "The global market for power tools – focus on industry and markets," *Business Economics*, July 2006, p.66-73.
- ³⁰ Hilti internal presentation on fleet management; Stefan Michel, "Hilti Fleet Management: Strategically Moving from Products to B2B Solutions," The Case Centre No. IMD-3-2354 (The Case Centre, 2013).
- ³¹ Hilti internal presentation on fleet management; Stefan Michel, "Hilti Fleet Management: Strategically Moving from Products to B2B Solutions," The Case Centre No. IMD-3-2354 (The Case Centre, 2013).
- ³² Arndt Paul, "Hilti Flottenmanagement – Serviceinnovation am Beispiel von Elektrowerkzeugen," in *Serviceinnovationen in Industrieunternehmen erfolgreich umsetzen*, edited by S. Schweiger, K. Dressel, and B. Pfeiffer, (Gabler: Wiesbaden, 2000).
- ³³ Interview with Arndt Paul, June 13, 2016.
- ³⁴ Arndt Paul, "Hilti Flottenmanagement – Serviceinnovation am Beispiel von Elektrowerkzeugen," in *Serviceinnovationen in Industrieunternehmen erfolgreich umsetzen*, edited by S. Schweiger, K. Dressel, and B. Pfeiffer, (Gabler: Wiesbaden, 2000).
- ³⁵ Arndt Paul, "Hilti Flottenmanagement – Serviceinnovation am Beispiel von Elektrowerkzeugen," in *Serviceinnovationen in Industrieunternehmen erfolgreich umsetzen*, edited by S. Schweiger, K. Dressel, and B. Pfeiffer, (Gabler: Wiesbaden, 2000).
- ³⁶ Arndt Paul, "Hilti Flottenmanagement – Serviceinnovation am Beispiel von Elektrowerkzeugen," in *Serviceinnovationen in Industrieunternehmen erfolgreich umsetzen*, edited by S. Schweiger, K. Dressel, and B. Pfeiffer, (Gabler: Wiesbaden, 2000).
- ³⁷ Stefan Michel, "Hilti Fleet Management: Strategically Moving from Products to B2B Solutions," The Case Centre No. IMD-3-2354 (The Case Centre, 2013).
- ³⁸ Hilti internal presentation on fleet management.
- ³⁹ Interview with Jan Doongaji, August 30, 2016.
- ⁴⁰ Interview with Jan Doongaji, August 30, 2016.
- ⁴¹ Mark W. Johnson, *Seizing the White Space – Business Model Innovation for Growth and Renewal* (Boston, MA: Harvard Business Press, 2010).
- ⁴² "Fleet Management: An Innovative Hilti Service for Innovative Customers: Winner of the Martin Hilti Innovation Prize," Hilti internal presentation, 2000, Schaan, Liechtenstein.
- ⁴³ Stefan Michel, "Hilti Fleet Management: Strategically Moving from Products to B2B Solutions," The Case Centre No. IMD-3-2354 (The Case Centre, 2013).